

EDUGENIE – GOOGLE GEMINI POWERED LEARNING ASSISTANT

PHASE 5: Project Development Phase

1. Introduction

The development phase transforms the approved design into a working EduGenie application or prototype. The development work follows the requirements and system design prepared in earlier phases.

2. Interface Development

The user interface is created with a simple structure. The user should be able to understand where to enter a question and where the generated answer will appear.

3. User Input

A question-input component is implemented so that students can enter educational requests using natural language. Input handling ensures that the submitted content reaches the processing stage.

4. Gemini Integration

Google Gemini is connected as the AI-powered component. The implementation sends suitable user input to the Gemini service through the selected integration method and receives the generated response.

5. Response Processing

After Gemini returns a response, the application handles the result and prepares it for presentation. The response is displayed in a clear format so that the user can read and understand it.

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PHASE 5: Project Development Phase – CONTINUED

6. Basic Validation

The system can check simple conditions such as whether the user has entered a question. Empty or invalid input should be handled with suitable feedback.

7. User Experience Improvements

During development, the team improves readability, navigation, response presentation, and general usability. Unnecessary interface elements are removed to keep the learning interaction focused.

8. Integration Testing During Development

Individual components are checked as they are developed. The team verifies that input handling, AI communication, response retrieval, and output display work together correctly.

9. Development Challenges

Potential challenges include API or service configuration, connectivity problems, response formatting, unexpected user input, and differences in AI-generated answers. These issues are addressed through repeated testing and corrections.

10. Conclusion

The development phase produces the working core of EduGenie. The completed workflow connects student input with Gemini-powered response generation and displays the result through a student-friendly interface.